FIITJEE

ALL INDIA TEST SERIES

JEE (Advanced)-2023

CONCEPT RECAPITULATION TEST – III

PAPER –1

TEST DATE: 26-04-2023

Time Allotted: 3 Hours **Maximum Marks: 180**

General Instructions:

- The test consists of total 54 questions.
- Each subject (PCM) has 18 questions.
- This question paper contains **Three Parts.**
- **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
- Each **Part** is further divided into **Two Sections**: **Section-A & Section-B**.

Section – A *(01 –06, 19 – 24, 37 – 42):* This section contains **EIGHTEEN (18)** questions. Each question has **FOUR** options. **ONE OR MORE THAN ONE** of these four option(s) is(are) correct answer(s).

Section – A *(07 – 10, 25 – 28, 43 – 46):* This section contains **TWELVE (12)** Matching List Type Questions. Each question has **FOUR** statements in **List-I** entries (I), (II), (III) and (IV) and **FIVE** statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which, **ONLY ONE** of these four options is correct answer.

Section – B *(11 – 18, 29 – 36, 47 – 54):* This section contains **TWENTY FOUR (24)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.

MARKING SCHEME

Section – A (One or More than One Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+4	If only (all) the correct option(s) is (are) chosen;
Partial Marks	:	+3	If all the four options are correct but ONLY three options are chosen;
Partial marks	:	+2	if three or more options are correct but ONLY two options are chosen and both of which are correct;
Partial Marks	:	+1	If two or more options are correct but ONLY one option is chosen and it is a correct option;
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–2	In all other cases.

Section – A (Single Correct): Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct option is chosen.
Zero Marks	:	0	If none of the options is chosen (i.e. the question is unanswered);
Negative Marks	:	–1	In all other cases.

Section – B: Answer to each question will be evaluated according to the following marking scheme:

Full Marks	:	+3	If ONLY the correct numerical value is entered at the designated place;
Zero Marks	:	0	In all other cases.

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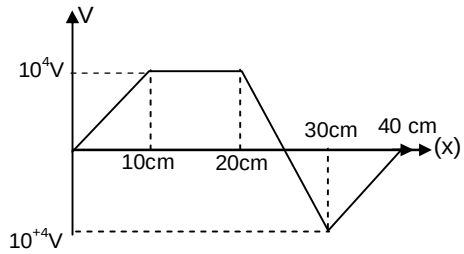
Physics

PART – I

Section – A (Maximum Marks: 24)

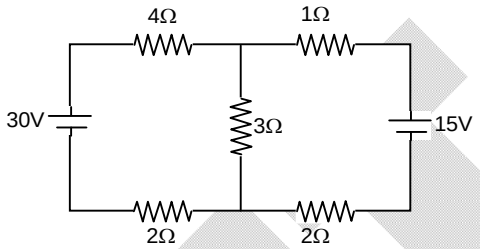
*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

1. For the given potential versus distance graph for one dimensional electric field and a particle of charge 20 μC and mass 0.2 kg. Choose the correct option(s):



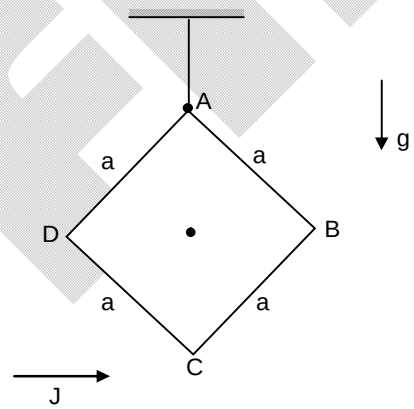
(A) force acting on the particle at x = 25 cm is zero
(B) Force acting on the particle at x = 25 cm is $4\hat{i}$
(C) minimum speed at origin required by the particle to reach x = 25 cm is 2 m/s
(D) minimum speed at origin required by the particle to reach x = 25 cm is $\sqrt{2}$ m/s

2. For the given circuit choose the correct option(s):



(A) Power delivered by 30 V battery is 90 W
(B) Power delivered by 15 V battery is 15 W
(C) current through 3 Ω resistor is 3 A
(D) Power loss in 1 Ω resistor is 1 W

3. A square plate of mass m and side length 'a' is suspended from an inextensible string. A horizontal impulse J is applied at lowest point C. Choose the correct option(s). Just after the application of impulse



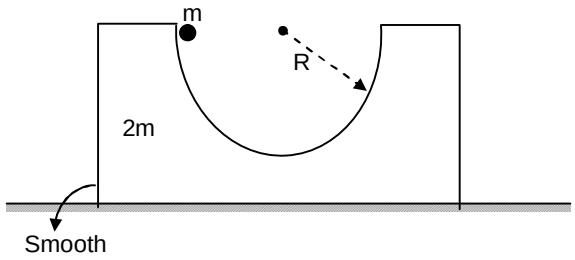
(A) Point A will move in forward direction
(B) Point A will move in backward direction

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(C) Instantaneous axis of rotation is $\frac{a}{3\sqrt{2}}$ distance above centre 'O'

(D) Rotational kinetic energy is more than translational kinetic energy

4. A wedge is cut in cylindrical shape. Its mass is 2m. A particle of mass m is kept at its edge and it released. All surfaces are smooth. Choose the correct option(s). When the particle reaches its lowest point. (R is radius)



(A) distance moved by wedge is R/3

(B) speed of particle is $2\sqrt{\frac{gR}{3}}$

(C) Normal reaction is 4 mg

(D) All of these

5. An ideal gas initially at pressure P_0 undergoes a free expansion (against vacuum) under adiabatic condition until its volume is 3 times its initial volume. The gas is next adiabatically compressed back to its original volume. The pressure after compression is $3^{2/3}P_0$. Then choose the correct option(s)

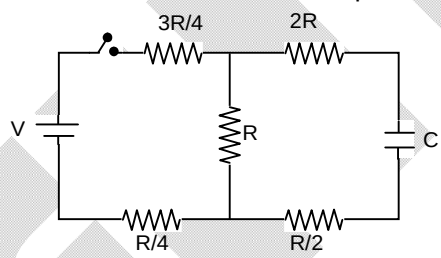
(A) The pressure of the gas after free expansion is $\frac{P_0}{3}$

(B) The pressure of the gas after free expansion is $P_0^{\frac{1}{3}}$

(C) The gas is monoatomic

(D) The gas is diatomic

6. Circuit is closed at t = 0 and capacitor was uncharged initially. Choose the correct option(s):



(A) charge on the capacitor is $q = \frac{CV}{2}\left(1 - e^{-\frac{t}{3RC}}\right)$

(B) charge on the capacitor is $q = \frac{CV}{2}\left(1 - e^{-\frac{2t}{5RC}}\right)$

(C) current in resistor R is $i = \frac{V}{2R}\left(1 - e^{-\frac{2t}{5RC}}\right)$

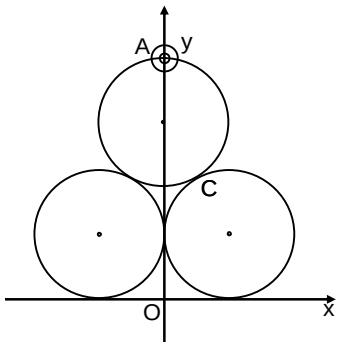
(D) current in resistor R is $i = \frac{V}{2R}\left(1 - e^{-\frac{t}{3RC}}\right)$

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Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

7. Consider three identical uniform disc of mass m and radius 'R'. System is kept in x-y plane match the moment of inertia about axis in List I to answer in List II:



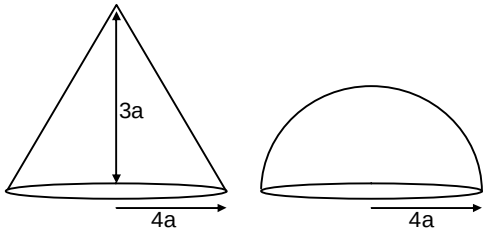
List - I		List - II	
(I)	About x axis	(P)	$\left(\frac{18+4\sqrt{3}}{4}\right)MR^2$
(II)	About y axis	(Q)	$\frac{11\,MR^2}{4}$
(III)	About an axis passing through centre of mass of system and perpendicular to its plane	(R)	$\frac{11\,MR^2}{2}$
(IV)	About axis passing through point A and perpendicular to plane	(S)	$\left(\frac{25+4\sqrt{3}}{2}\right)MR^2$
		(T)	$\left(\frac{27+8\sqrt{3}}{4}\right)MR^2$

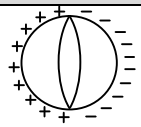
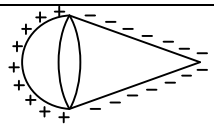
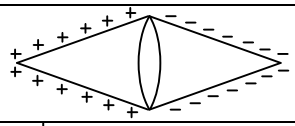
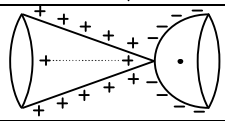
Codes:

	I	II	III	IV
(A)	T	Q	R	S
(B)	P	Q	R	S
(C)	S	R	Q	T
(D)	P	Q	S	R

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8. Consider a hollow cone of radius 4a and height 3a, and a hemisphere shell of radius 4a. Various combinations of them are made by giving them equal and opposite charge Q, Match their dipole moment given in List II.

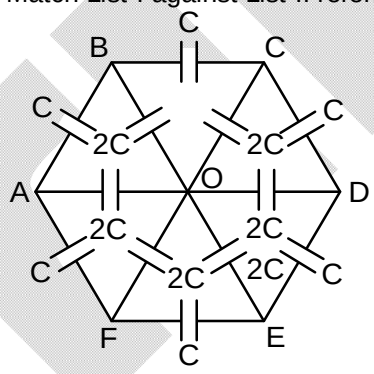


List - I		List - II	
(I)		(P)	4Qa
(II)		(Q)	5Qa
(III)		(R)	2Qa
(IV)		(S)	3Qa
		(T)	Zero

Codes:

	I	II	III	IV
(A)	P	R	Q	P
(B)	T	Q	R	S
(C)	P	Q	R	P
(D)	Q	R	T	P

9. Match List-I against List-II referring to the given figure.



List - I		List - II	
(I)	Effective capacitance between A and D	(P)	$\frac{45}{13}$C
(II)	Effective capacitance between F and E	(Q)	$\frac{15}{8}$C

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(III)	Effective capacitance between A and E	(R)	$\frac{45}{19}$C
(IV)	Effective capacitance between A and O	(S)	$\frac{9}{5}$C
		(T)	$\frac{3}{5}$C

Codes:

	I	II	III	IV
(A)	P	T	R	S
(B)	Q	P	R	S
(C)	S	R	Q	P
(D)	S	Q	R	T

10. List-I shows position vector as function of time for a particle. Match the characteristic of the motion with their description in List-II:

List - I		List - II	
(I)	A particle has its position as a function of time given by $\vec{r}(t)=\vec{r}_0+\vec{v}_0t$ where $\vec{r}_0$ and $\vec{v}_0$ are non-zero constants.	(P)	Angular momentum of the system about the origin is conserved.
(II)	A particle has its position vector as function of time given by $\vec{r}(t)=v_0t\hat{k}+(a_0\cos\omega t)\hat{i}+(a_0\sin\omega t)\hat{j}$ where $\vec{a}_0$ and $\vec{v}_0$ are non-zero constant. (angle between $\vec{v}_0$ and $\vec{a}_0$ is $90°<\theta<180°$)	(Q)	The particle may be charged and moving in a uniform magnetic field, no other force acts on it.
(III)	A particle is moving in a plane with its position vector given by $\vec{r}(t)=(a_0\sin\omega t)\hat{i}+(b_0\cos\omega t)\hat{j}$ where a_0, b_0 $(a_0\neq b_0)$ and ω are positive constant.	(R)	Projection of particle on x or y axis is simple harmonic motion.
(IV)	A particle is moving in a plane with position vector $\vec{r}(t)=(a_0\cos\omega t)\hat{i}+(a_0\sin\omega t)\hat{j}$ where a_0 and ω are positive constants.	(S)	Kinetic energy of the particle is constant.
		(T)	Area swept by position vector of particle in equal time interval is equal.

Codes:

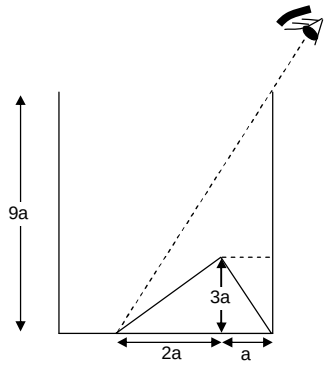
	I	II	III	IV
(A)	PQST	PQRST	R	PQRST
(B)	STPQ	QPRST	S	PQRS
(C)	PQRS	STPQR	PQ	RSPQT
(D)	PQRST	STPQ	RS	PQRS

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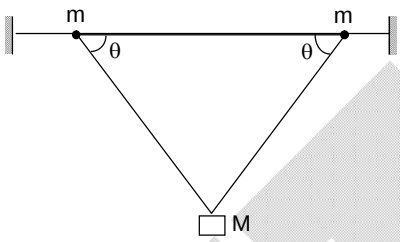
Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

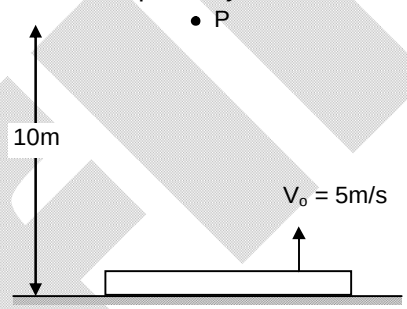
11. In a container of depth '9a' a triangular wedge is kept touching a wall as shown such that its one edge is just visible to an external observer. When a liquid of refractive index μ is completely filled in the container top of wedge also becomes visible. Find the retractive index of liquid.



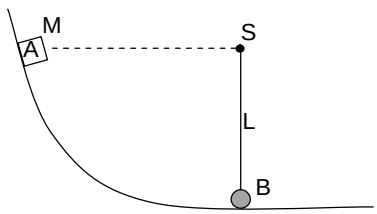
12. Two rings which can slide along rough rod (μ = 0.3) are supporting a block of mass M = 3kg mass of each ring is m = 1 kg. What is maximum value of cot θ, for the system to remain in equilibrium?



13. A light particle P is dropped at t = 0 from height 10 m. At the same time a heavy plank starts moving upward with constant velocity $v_0 = 5\,m/s$. Find maximum height reached by ball from ground after perfectly elastic collision with plank. (Give answer in meters)

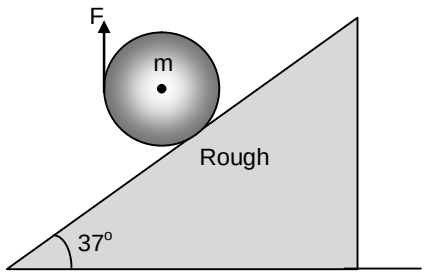


14. A bob B of mass m is suspended by an extensible string of length ℓ. Another block A of mass M is released from the same level as point of suspension. After perfectly elastic collision bob B just completes vertical circle about point S. Find ratio of M/m.
(Given : $\sqrt{5} = 2.22$, $\sqrt{2} = 1.41$)



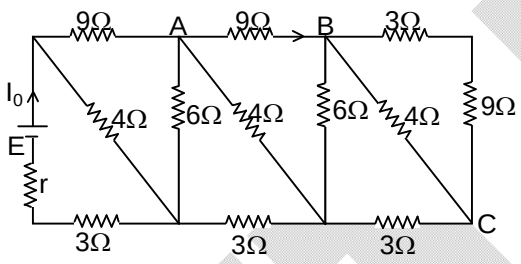
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15. A solid sphere is at rest on a rough inclined plane as shown in figure. The force required to keep the sphere at rest is F. Find the value of $\frac{2F}{mg}$.



16. A nucleus ${}_3x^7$ has binding energy/nucleon of 10 MeV. It absorbs a proton and its mass increases by $\frac{99}{100}$ times mass of proton. The new binding energy of the nucleus is given by E. Find the value of E (in MeV) (Take 1 amu = 930 MeV)

17. In the circuit shown if current flowing through battery is I and current in branch AB is I_1. Find the ratio of $\frac{I_0}{I_1}$.



18. Consider the following decay

$${}^{236}_{92}U \longrightarrow {}^{232}_{90}Tb + X$$

Mass

$\left({}^{236}_{92}U = 236.045562 \text{ amu},\ {}^{232}_{90}Th = 232.038054 \text{ amu},\ {}_0n^1 = 1.008665 \text{ amu},\ {}^1_1H = 1.007277 \text{ amu},\right.$

$\left. {}^4_2He = 4.002603 \text{ amu and } 1 \text{ amu} = 1.5 \times 10^{-10} \text{ J} \right)$

If amount of energy decay in this reaction is $n \times 10^{-12}$ J, find n. (Round off to two decimal places)

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Chemistry

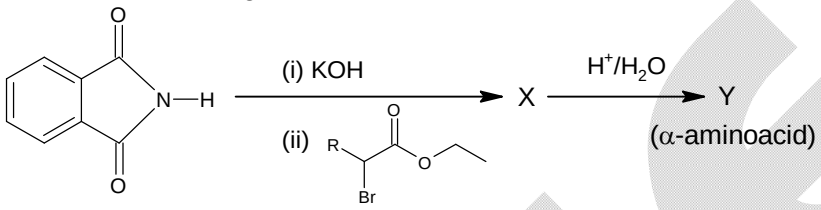
PART – II

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

19. Pick the correct statement(s)
(A) Number of carboxylic acid groups present in tartaric acid is two.
(B) The IUPAC name of cinnamic acid is 2-phenyl but-2-enoic acid
(C) Prop-2-enoic acid is also known as crotonic acid
(D) The IUPAC name of X is propane-1,2,3-trinitrile, where X is $NC-CH_2-CH(CN)-CH_2-CN$

20. Consider the following reaction



Now pick the right statement(s):
(A) Above synthesis is known as Gabriel phthalimide synthesis
(B) X contain two nitrogen atom per molecule
(C) If $R=(CH_3)_2CH-$, Y is leucine
(D) If R = H, Y will be optically inactive

21. Consider $P(CH_3)_3(CF_3)_2$ and pick correct statement(s):
(A) Maximum number of atoms that may lie in one plane is 7.
(B) Four planes are possible with 7 atoms in each plane
(C) $-CH_3$ occupy equatorial positions
(D) $-CF_3$ occupy axial position

22. The K_p value for the reaction; $A_{2(g)}+B_{2(g)} \rightleftharpoons 2AB_{(g)}$ at 500^oC is 64. If the initial pressure of A_2 and B_2 is 0.6 atm each, then
(A) p_{A_2} at equilibrium = 0.12 atm
(B) p_{B_2} at equilibrium = 0.12 atm
(C) p_{AB} at equilibrium = 0.48 atm
(D) p_{AB} at equilibrium = 0.24 atm

23. Which of the following is ore of copper?
(A) Bauxite
(B) Cuprite
(C) Malachite
(D) Azurite

24. Pick the carbonates which are white in colour.
(A) $BaCO_3$
(B) $CuCO_3$
(C) $CaCO_3$
(D) $PbCO_3$

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Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

25. Match List-I with List-II:

List – I (Reaction)		List – II (Product)	
(I)	Aldol condensation	(P)	Primary amine
(II)	Hoffmann bromamide degradation	(Q)	Esters
(III)	Baeyer – Villiger oxidation	(R)	α,β unsaturated aldehyde
(IV)	Hell-Volhard Zelinsky reaction	(S)	α -halo carboxylic acids
		(T)	β-keto ester

Codes:

	I	II	III	IV
(A)	S	P	Q	T
(B)	RS	P	Q	T
(C)	RS	S	T	S
(D)	R	P	Q	S

26. Match List-I with List-II:

List – I (Reaction)		List – II (Major product)	
(I)	Wurtz reaction	(P)	Anti Markovnikov (alcohol)
(II)	Corey – house synthesis	(Q)	Markovnikov (alcohol)
(III)	Propene $\xrightarrow{HBO}$	(R)	Unsymmetrical aldehyde
(IV)	Propene $\xrightarrow{OMDM}$	(S)	Symmetrical alkane
		(T)	Unsymmetrical alkane

Codes:

	I	II	III	IV
(A)	S	S	P	Q
(B)	T	R	Q	P
(C)	S	ST	P	Q
(D)	T	S	Q	P

27. Match List-I with List-II:

List - I		List - II	
(I)	Mg^{2+}	(P)	Paramagnetic species
(II)	Zn^{2+}	(Q)	Diamagnetic species
(III)	Mn^{2+}	(R)	Spin multiplicity = 0
(IV)	Fe^{2+}	(S)	Magnetic moment = 0
		(T)	Magnetic moment = 5.92 B.M

Codes:

	I	II	III	IV
(A)	QR	QR	PT	P
(B)	QS	QS	PT	PT
(C)	QRS	QR	PT	PT
(D)	QRS	QRS	PT	P

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28. Match List-I with List-II:

List - I		List - II	
(I)	Isoelectronic species	(P)	Astatine
(II)	f-block element	(Q)	Rhenium
(III)	d-block element	(R)	Ne, Mg^{2+}
(IV)	p-block element	(S)	Terbium
		(T)	N^{3-}, F^-

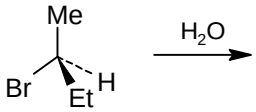
Codes:

	I	II	III	IV
(A)	RT	Q	S	P
(B)	RT	Q	P	S
(C)	RT	S	Q	P
(D)	RT	S	P	Q

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

29. Find the percentage of R-configuration in product, if 92.46% racemization takes place for given optically pure compound.



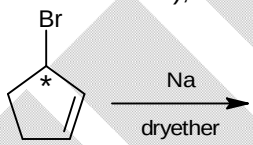
30. Consider following reaction,

$HCHO + NH_3 \longrightarrow X$

(excess) x mole (1 mole, compound used in urinary infection)

Find $\left(\frac{x}{100}\right)$ assuming 100% yield.

31. How many products having ten carbon atoms are obtained in following reaction (excluding stereoisomers),



(Note: Asterik (*) represent labelled carbon)

32. Calculate enthalpy change (kJ/mole) of the following reaction:

$H_2C = CH_{2(g)} + H_{2(g)} \longrightarrow H_3C - CH_{3(g)}$

Take bond energy of $C-H$, $C-C$, $C=C$, $H-H$ as 410, 350, 620 and 440 kJ/mole, respectively.

33. If equilibrium constant for below reaction is $x \times 10^y$ (written in scientific notation). Then find $(y-x)$.

$A^{2+}_{(aq)} + B^{4+}_{(aq)} \rightleftharpoons A^{3+}_{(aq)} + B^{3+}_{(aq)}$

Given, $E^0_{A^{3+}/A^{2+}} = 0.8V$, $E^0_{B^{4+}/B^{3+}} = 1.39$ V and $\frac{2.303RT}{F} = 0.059$

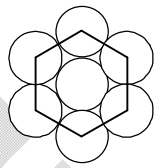
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34. The number of fluorine atoms in compound (X) in below reaction,

$$\underset{\text{(excess)}}{Xe_{(g)}} + F_{2(g)} \xrightarrow[1\text{ bar}]{673\text{ K}} X \text{, is}$$

35. The number of N–B–N linkage in borazole is

36. The packing efficiency in percentage of the two dimensional hexagonal unit cell shown below is: (Take $\frac{\pi}{\sqrt{3}} = 1.8144$)



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Mathematics

PART – III

Section – A (Maximum Marks: 24)

*This section contains **SIX (06)** questions. Each question has **FOUR** options (A), (B), (C) and (D). **ONE OR MORE THAN ONE** of these four option(s) is (are) correct answer(s).*

37. If $a_1, a_2, \ldots\ldots\ldots, a_n$ is a sequence of positive numbers which are in A.P. with common difference d and $a_1 + a_4 + a_7 + \ldots\ldots\ldots + a_{16} = 147$, then $a_1 + a_{16} = M$ and $a_1 + a_6 + a_{11} + a_{16} = N$ and maximum value of $a_1 a_2 a_3 \ldots\ldots\ldots a_{16} = \left(\frac{S}{W}\right)^{16}$ (where S and W are coprime), then

(A) M = 49
(B) N = 98
(C) S = 49
(D) W = 4

38. Let $A(5,-1), B(\alpha,-7)$ and $C(-2,\beta)$ are vertices of ΔABC and (–6, –4) is the image of orthocentre of ΔABC in the point mirror M which is mid-point of the side BC. If (p, q) is circumcentre of ΔABC, then

(A) the value of $\beta^2 - \alpha^2 + 5\beta - \alpha$ is 12
(B) the value of (2p + 1) is zero
(C) the value of (2q + 5) is –6
(D) the value of $q^2 - \frac{p}{2}$ is $\frac{13}{2}$

39. The value(s) of 'a' for which equation $\frac{x^3 - 6x^2 + 11x - 6}{x^3 + x^2 - 10x + 8} + \frac{a}{30} = 0$ does not have real solution is/are

(A) –30
(B) –10
(C) 5
(D) 12

40. A polynomial function f(x) with non-negative coefficient satisfy the equation $f(f(x)) = x\int_0^x f(t)dt$ and f(0) = 0, then

(A) number of points where $|f(|x|)|$ is non derivable is 0
(B) sgn(f(x)) is discontinuous at x = 1
(C) derivative of f(x) w.r.t $\sin^{-1}\left(\frac{2x}{x^2+1}\right)$ at $x = \sqrt{3}$ is –4
(D) $\lim_{x\to 0^+}\left(\frac{\sqrt{3}f(x)}{x}\right)^x = 1$

41. Let Q (a, b, c) is a variable point on the plane x – y + z = 3 and A(0, 0, 0), B(1, 1, 1) are any other two points in the space then identify the correct statement(s)

(A) If QA + QB is minimum then a – c – 10b is equal to 1
(B) If QA + QB is minimum then a – c – 10b is equal to 2
(C) If $|QA - QB|$ is maximum then $a - 2b + c$ is equal to 0
(D) If $|QA - QB|$ is maximum then $a - 2b + c$ is equal to 2

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42. If a complex number z satisfies relation $\arg\left(z^2-w\right)=\frac{\pi}{2}+\arg\left(z^2-w^2\right)$ where $w=e^{i\frac{2\pi}{3}}$, then identify the correct statement(s):

(A) Maximum value of |z| is $\frac{\sqrt{3+\sqrt{3}}}{2}$

(B) Maximum value of |z| is $\frac{\sqrt{2+2\sqrt{3}}}{2}$

(C) Maximum value of $|z-1-i||z+1+i|$ is equal to $\frac{\sqrt{17}+\sqrt{3}}{2}$

(D) Minimum value of $|z-1-i||z+1+i|$ is equal to $\frac{\sqrt{17}-\sqrt{3}}{2}$

Section – A (Maximum Marks: 12)

*This section contains **FOUR (04)** Matching List Type Questions. Each question has FOUR statements in **List-I** entries (I), (II), (III) and (IV) and FIVE statements in **List-II** entries (P), (Q), (R), (S) and (T). The codes for lists have choices (A), (B), (C), (D) out of which **ONLY ONE** of these four options is correct answer.*

43. Match List-I with List-II:

List - I		List - II	
(I)	If z is a complex number such that $\text{Im}\left(z^2\right)=3$, then eccentricity of the locus is	(P)	$\sqrt{3}$
(II)	If the latus rectum of a hyperbola through one focus subtends 60^0 angle at the other focus, then its eccentricity is	(Q)	3
(III)	If A (3, 0) and B (–3, 0) and locus of P which is hyperbola is given by the condition $PA-PB=4$, then eccentricity of the conjugate hyperbola is	(R)	$\sqrt{2}$
(IV)	If the angle between the asymptotes of hyperbola is $\frac{\pi}{3}$, then the eccentricity of its conjugate hyperbola is	(S)	$\frac{3}{\sqrt{5}}$
		(T)	2

Codes:

	I	II	III	IV
(A)	T	S	P	R
(B)	R	P	S	T
(C)	P	R	T	S
(D)	T	S	P	T

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44. If a function $f(x)=\sin^{-1}(2x-1)+\cos^{-1}\left(2\sqrt{x-x^2}\right)+\tan^{-1}\left(\frac{1}{1+\left[x^2\right]}\right)$ (where [.] represents greatest integer function), then

List - I		List - II	
(I)	$f\left(\frac{1}{6}\right)$ is equal to	(P)	$\frac{\pi}{6}$
(II)	$f\left(\frac{3}{4}\right)$ is equal to	(Q)	$\frac{\pi}{4}$
(III)	$\sin^{-1}(\tan(f(1)))$ is equal to	(R)	$\frac{\pi}{3}$
(IV)	$\sum_{r=1}^{10} f\left(\frac{r}{20}\right)$ is equal to	(S)	$\frac{7\pi}{12}$
		(T)	$\frac{5\pi}{2}$

Codes:

	I	II	III	IV
(A)	Q	S	P	R
(B)	Q	S	P	T
(C)	T	S	P	R
(D)	T	S	P	T

45. Match List-I with List-II:

List - I		List - II	
(I)	If a plane meets the coordinate axes in A, B, C and centroid of ΔABC is $\left(\frac{1}{3}, \frac{2}{3}, 1\right)$ then equation of plane is	(P)	$12x-99y+3z-28=0$
(II)	The equation of the plane passing through the intersection of the plane $2x-y+z-6=0$ and $x+y+2z-7=0$ and perpendicular to the plane $2x+3y+z=0$ is	(Q)	$x+y+z=0$
(III)	The equation of the plane containing the lines $\frac{x-3}{2}=\frac{y+1}{-3}=\frac{z+2}{1}$ and $\frac{x-7}{-3}=\frac{y}{1}=\frac{z+7}{2}$ is	(R)	$6x+3y+2z=6$
(IV)	The equation of the plane parallel to the lines $\frac{x-1}{2}=\frac{y+1}{3}=\frac{z-7}{4}$ and $\frac{x-3}{1}=\frac{2y-9}{4}=\frac{z}{1}$ and passing through point (1, –1, 2) is	(S)	$x+y+z+3=0$
		(T)	$5x-2y-z=5$

Codes:

	I	II	III	IV
(A)	P	T	P	R
(B)	Q	P	T	R
(C)	R	P	Q	T
(D)	T	Q	P	R

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46. Match List-I with List-II:

List - I		List - II	
(I)	Number of integral values of x satisfying the inequality $\left(\frac{3}{4}\right)^{6x+10-x^2} < \frac{27}{64}$ is	(P)	2
(II)	Number of positive integers x for which $f(x) = x^3 - 8x^2 + 20x - 13$, is a prime number, is	(Q)	3
(III)	Total number of integral values of 'a' such that $x^2 + ax + a + 1 = 0$ has integral roots is equal to	(R)	4
(IV)	In a ΔABC, BC = 8, CA = 6 and AB = 10. A line dividing the triangle ABC into two regions of equal area is perpendicular to AB at the point X, then the value of $\frac{BX}{\sqrt{2}}$ is	(S)	8
		(T)	7

Codes:

	I	II	III	IV
(A)	R	P	Q	T
(B)	Q	P	R	T
(C)	P	Q	T	R
(D)	T	Q	P	R

Section – B (Maximum Marks: 24)

*This section contains **EIGHT (08)** numerical based questions. The answer to each question is a **NUMERICAL VALUE.** If the numerical value has more than two decimal places, truncate/round-off the value to **TWO** decimal places.*

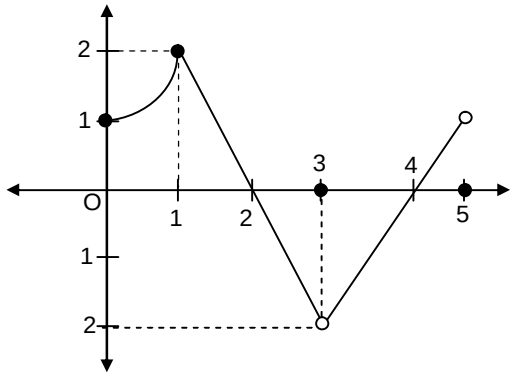
47. Let the hyperbola $H: x^2 + y^2 + 4xy + 8x + 8y + 8 = 0$ and the line $L: x + y + 1 = 0$ intersects in two distinct points. If radius of the circle which touches the hyperbola at the points where H meets the line L is R, then the value of R^2 is……

48. Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ and $B = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix}$ where $a_{ij} \forall i,j$ are natural numbers. If $\text{trace}\left(AB + AB^3 + AB^5 + \cdots\ldots + AB^{19}\right) = 210$, then the number of ordered triplets $(a_{31} + a_{22} + a_{13})$ is……..

49. The roots of the polynomial equation $x^9 - 6x^8 + 12x^7 - 8x^6 = 0$ form a scalar matrix $A = \left[a_{ij}\right]_{3\times 3}$ and matrix $B = \text{diag}(b_{11}, b_{22}, b_{33})$ is a diagonal matrix order 3 × 3 where $b_{rr} = \sum_{j=1}^{3}\sum_{i=1}^{3} a_{ij}\ \forall r = 1,2,3$. If $\frac{|\text{adj}B|}{|\text{adj}(\text{adj}(\text{adj}A))|} = \left(\frac{p}{q}\right)^6$, where p, q are relatively prime number, then the value of (2q – p) is_____.

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50. Graph of a function y = f(x) is shown as



If $g(x) = \left|f\left(|x|\right)\right|$, then the number of solution(s) of the equation $g\left(g(x)\right) = \text{sgn}\left(x^2 - (k+1)x + (k^2+1)\right)$, $k \in R$ is_____.

51. Let f : R $\to$ R be a continuous function which satisfies $f(2x) = 3f(x)\ \forall x \in R$. If $\int_0^1 f(x)dx = 1$, then the value of $\frac{1}{2}\int_1^2 f(x)dx$ is_____.

52. Let Z (Z $\in$ complex number) be one of the roots of the equation $x^2 - \left(\log_2 \alpha - \log_2 \beta\right)x + \cos\alpha - \sin\beta = 0$. If the harmonic mean of the roots is 2 and |Z| = 1, then the sum of all values of β in degrees when $0 < \beta < 360^0$ is _____.

53. Consider $f(x) = x^4 + ax^3 + bx^2 + cx + d$. If straight line y = 3x + 2 is tangent to the curve y = f(x) at $P\left(1,\ f(1)\right)$ which again intersects f(x) at Q(2, 8) and $f''(1) = 0$ then the value of f(3) is _____.

54. Let f be a differentiable function defined in [0, 1] such that $f(f(x)) = x$ and $f(0) = 1$. If the value of $\int_0^1 \left(x - f(x)\right)^{2022} dx = \frac{p}{q}$ where p and q are relatively prime then the value of (p + q) is _____.

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